

Detecting, mapping and digitising canopy geometry, fruit number and peel colour in pear trees with different architecture

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ABSTRACT

Increasing farm labour and input costs and the requirement for more orchard data are leading to rapid advances in technology to improve management systems in fruit production. The study aimed to (i) validate a sensorised platform to estimate fruit number, peel colour and blush coverage in pear orchards with different pear selections and tree architectures, (ii) establish relationships between fruit number, peel colour and canopy geometry features, and (iii) evaluate the platform for mapping and digitising orchard features. The study was carried out over two years in an experimental 'ANP-0131' orchard and in one year in two commercial pear orchards ('ANP-0131' and 'PremP009'). Predictions of fruit number and blush coverage were compared in traditional three-dimensional (3D) and modern high-density two-dimensional (2D) training systems. Overall, prediction errors for fruit number were < 6.5 % in all the training systems, but improved performance was achieved in vertical 2D configurations (% standard errors = 2.2 %). Fruit number and blush coverage per unit of leaf area were higher in 2D compared to 3D training systems. Blush coverage predictions in 'ANP-0131' were reliable ($R^2 = 0.67$, RMSE = 3.70 %). Accurate predictions of blush coverage classes were achieved by modifying sample variance. Fruit number and blush coverage were negatively affected by increasing canopy size. The platform proved useful for mapping and digitisation. Spatial heatmaps of orchard features provided a valuable visual aid to identify zones for priority interventions for peel colour enhancement.

1. Introduction

Current and future developments of precision orchard management require incremental technological uptake within farms. Obtaining accurate and scalable data from the orchard should be a prerequisite for fruit growers to accept and incorporate technology in their business models. Orchard operations such as crop load and fruit quality management often rely on small subsamples of fruit or subjective visual observations of fruit size, number and colour carried out by seasonal workers with limited experience. In addition to the low-sample and high-subjectivity limitations, the cost of farm labour is increasing worldwide, leading to Australian fruit growers struggling to maintain profitability (Amin-Chaudhry et al., 2022). It is therefore pivotal to reduce orchard costs by, amongst others, increasing technology adoption and replacing expensive untrained seasonal workers with the long-term goal to maintain fruit production sustainability in the region.

Technology to detect fruit in the orchard is now widely available at experimental, prototype-release, and commercial stages, and it represents a large topic in horticulture research (Ukwuoma et al., 2022). Currently, the majority of technology used for fruit detection in orchards rely on the integration of machine vision and deep learning algorithms that can often achieve accuracy over 95 % in experimental conditions (Anderson et al., 2021; Koirala et al., 2019). Detection errors are typically caused by factors such as changes in illumination, foliage occlusions, fruit overlaps and next-row detections (Tang et al., 2023). As a consequence, due to better visibility, fruit detection accuracy in tree canopies with two-dimensional (2D) configurations improve compared to three-dimensional (3D) systems, as demonstrated in apple (Bortolotti et al., 2021) and stone fruit (Islam et al., 2022). Accurate estimates of fruit number at different stages of the fruit growing season—i.e., from fruit set to harvest—can empower fruit growers to take decisions for fruit thinning requirements (e.g., labour for hand thinning; type of

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chemical thinning agent, rates and number of applications), yield estimation and canopy management (pruning / thinning cuts) to regulate productivity and mitigate biennial bearing.

Tree canopy architecture parameters can be measured using different technologies. Although attempts to accurately reproduce 3D vegetation structures with machine vision and ultrasounds have previously been carried out (Colaço et al., 2018; Kolhar and Jagtap, 2023; Zaman and Schumann, 2005), Light Detection and Ranging (LiDAR) generates significantly improved accuracy (Dandois and Ellis, 2010) and has become a widely accepted technology in horticultural industries. Important architectural parameters such as canopy height, width, density, volume and area, leaf area and leaf area index can be derived from LiDAR-3D cloud point imagery by applying geometrical concepts to the voxels (Escolà et al., 2017; Rosell Polo et al., 2009; Sanz-Cortiella et al., 2011; Sanz et al., 2013; Underwood et al., 2016). In addition, it was demonstrated that canopy light interception can be estimated or modelled from LiDAR-derived canopy geometry features (Örn, 2016; Scalisi et al., 2021a; Westling et al., 2018, 2020).

The Australian pear industry is currently threatened by concomitant increasing costs (mostly labour) and reduced domestic demand (Australian Agriculture Equities, 2021). To counter reduced demand, breeding programs and marketing campaigns need to target selections that produce fruit that are appealing for consumers and withstand long cold chain storage, while adapting to modern high-density orchards that maximise productivity and fruit quality. Worldwide, breeding programs have recently favoured the selection of pear fruit with red pigmentation (blush or full coverage) over traditional green pears. For example, an Australian breeding program recently developed and patented the two blush pear cultivars ‘ANP-0131’ (Turpin, 2015a) and ‘ANP-0118’ (Turpin, 2015b) that are currently marketed as Ricó™ and Lanya™, respectively, while a New Zealand breeding program patented the fully red cultivar ‘Prem009’ (White, 2015) registered as Piqa® Boo®.

Technology capable of block scale estimations of crop load, peel redness and blush coverage, fruit size and canopy size could enhance the implementation of precision management strategies to improve overall marketable yield, reduce within-orchard variability, and reduce costs associated with labour and manual operations. However, to date, there is no clear evidence that shows the reliability of remote or proximal sensing to accurately, and consistently, predict tree features at large scale in pear orchards with diverse training system configurations. A successful attempt to measure spatial variability of yield and tree vigour in a pear orchard was previously carried out by Perry et al. (2010) using UAV imagery. The accuracy of a commercial sensorised platform for predictions of (i) tree geometry, light interception, flower and fruit number in apple (Scalisi et al., 2021a), (ii) fruit number, size and colour in stone fruit (Islam et al., 2022), and (iii) fruit size in pears (Scalisi et al., 2023b) has been recently validated. Reliable digitisation of orchard features will allow in-depth understanding of temporal crop responses to numerous factors and is paramount to support the growth of orchard profitability and maintain long term sustainability.

Previous research unveiled underlying relationship trends between fruit number and peel colour development in pear cultivars (Scalisi et al., 2023a). The blush redness of ‘ANP-0131’ was reduced in larger fruit and in trees with large canopies, whereas colour development in full-red ‘PremP009’ pears was only marginally affected by fruit size and canopy size, and was likely to be more influenced by air temperature. This might explain why pear growers in the Goulburn Valley (Australia) reported lack of peel redness development in ‘PremP009’, as opposed to growers in cooler regions (New Zealand or Northern Italy). This phenomenon can also be explained by the different light environments in the growing regions. Some effects of temperature on the blush colour of ‘ANP-0118’ pears have been recently reported (McClymont et al., 2023).

Relationships between canopy parameters, crop load and fruit quality need to be further studied in trees with different canopy architectures to generate effective orchard management practice guidelines for growers. This study aimed to determine the reliability of a sensorised

platform to estimate fruit number and peel colour in pear orchards with different training system configurations, and to study their relationship with canopy geometry features. The research attempted to compare prediction reliability in traditional 3D and in modern high-density 2D training systems. The hypothesis was that machine vision performance for fruit detection and their feature estimation deteriorates in 3D canopy scans compared to 2D systems due to missing detections. In addition, the research aimed to determine the accuracy of blush coverage classification in a modern blush pear to inform supply chain and forecast packout, and to evaluate the use of the platform for mapping and digitisation of orchard features for precision agriculture applications.

2. Materials and methods

2.1. Orchards and experimental design

The study was carried out over the two summer seasons 2020–21 (year 1) and 2021–22 (year 2) in the Goulburn Valley (Greater Shepparton, Victoria, Australia). Greater Shepparton is the largest pear production region in Australia (72.9 %) (Australian Bureau of Statistics, 2022).

2.1.1. Experimental orchard

An experimental ‘ANP-0131’ pear orchard located at the Tatura SmartFarm (36°, 26′, 19.97″ S and 145°, 15′, 54.04″ E, 114 m above sea level) was used to obtain measurements in years 1 and 2. The orchard was planted in 2013 and consisted of three traditional training systems—i.e., spindle (SP), central leader (CL) and vase (VA)—and two modern high-density systems: a multileader vertical trellis (MLVT), and a multileader open Tatura trellis (MLOT). The orchard has 4.5 m row spacing and various tree spacings. Both MLVT and MLOT trees consisted of two horizontal cordons that had leaders (i.e., uprights) with 50 cm spacing, and equivalent to ~ 4444 and ~ 8889 leaders/ha for MLVT and MLOT, respectively. SP, CL, VA and MLOT systems were considered to have 3D canopies, whereas MLVT systems were classified as 2D fruiting walls.

The experiment was designed as a split-plot randomised complete block with three replicates of each treatment, except for MLOT and MLVT, which had 9 replicates. Each plot was 14 m along the tree row (north–south) with a central measurement row and two guard rows (east and west). Orchard management operations such as soil management, irrigation, nutrition, integrated pest and disease management and weed management followed commercial practices. Additional information on the experimental conditions at the Tatura SmartFarm is available in recent literature (McClymont et al., 2021).

2.1.2. Commercial orchards

Two commercial orchards (Calimna Orchard, Ardmona, Victoria, Australia) were used for technology validation during Year 2. The two blocks had different modern cultivars—‘ANP-0131’ and ‘PremP009’.

Table 1

Characteristics of commercial orchards (Calimna Orchard, Ardmona, Victoria, Australia) scanned in 2021–22 (year 2).

Orchard feature	Cultivar	
	‘PremP009’	‘ANP-0131’
Area (ha)	4.4	4.4
Tree spacing (m)	1.2	1.5
Row spacing (m)	4	4
Planting density (trees/ha)	2083	1667
Tree architecture	Tatura Trellis (Y)	Tatura Trellis (Y)
Leaders/ha	8332	10,002
Leaders/tree	4	6
Scan date	Pre-harvest (07 Feb 2022)	Pre-harvest (22 Feb 2022)

Commercial orchard characteristics are shown in Table 1.

2.1.3. Cultivar description

This study involved the use of the two modern cultivars ‘ANP-0131’ (*Pyrus communis*) and ‘PremP009’. ‘ANP-0131’ (*Pyrus communis*) was developed and patented in Australia (Turpin, 2015a). ‘ANP-0131’ is a bicolour pear with a green background and a dark-red overcolour (blush) in the peel. ‘ANP-0131’ was derived from crossing ‘Corella’ (seed parent) and ‘Doyenne du Comice’ (pollen parent) and is marketed with the name Ricó™. According to the Australian Apple and Pear industry (Apple and Pear Australia Ltd), blush coverage (%) represents one of the most important fruit quality characteristics to determine packout of Ricó™ fruit.

‘PremP009’ is an interspecific pear (*Pyrus communis* × *Pyrus pyrifolia* × *Pyrus bretschneideri*) developed and patented in a New Zealand breeding program (White, 2015). ‘PremP009’ has uniform dark-red colouration and is commercialised with the name Piqua® Boo®. The intensity of red colouration represents an important quality parameter for Piqua® Boo® pears.

Both ‘ANP-0131’ and ‘PremP009’ are under industry evaluation for large-scale production in order to meet the high demand from the south-east Asian market.

2.2. Scanning platform and technology

The sensorised mobile platform *Cartographer* (Green Atlas, Alexandria, NSW, Australia) was utilised for the collection of raw RGB images and to apply deep-learning models for the prediction of fruit number and colour. Concomitantly, light detection and ranging (LiDAR) point cloud images were logged by the platform to generate canopy architecture parameters. The platform utilised in this study included two RGB cameras, four custom-built strobe lights for canopy illumination, a LiDAR scanner (orientated to scan the vertical profile of the trees while passing), and a global navigation satellite system (GNSS). The reliability of this technology was recently validated on apples (Scalisi et al., 2021a) and stone fruit (Islam et al., 2022).

A phone-interface was used to control logging and record metadata to aid retrospective identification. Logging was switched on a few metres prior to the start of the measurement section and off a few metres past the end of the measurement section. Measurements were done early

in the morning—07:00 am Australian Eastern Daylight Time (AEDT)—in line with a previous study on stone fruit (Scalisi et al., 2022) that demonstrated that early morning or late afternoon scans can achieve statistically similar fruit colour estimates to nocturnal (ideal) measurements.

Fig. 1 shows examples of tree shape transects of the five systems under study (i.e., SP, CL, VA, MLVT and MLOT) obtained with LiDAR scanner mounted on the scanning platform. The orchard had retractable overhead hail netting, as shown in blue (Fig. 1).

The scans obtained in the ‘ANP-0131’ experimental site at the Tatura SmartFarm were done driving *Cartographer* at an average speed of 4.4 and 8.2 km h⁻¹ in 2021 and 2022, respectively. The low speed utilised in the first season was precautionary due to limited experience. The driving speed determined the final image sample size, as *Cartographer* logged images at an interval of 5 images/s. The sample size generated with 2022 scans was considered adequate for this experiment ($n > 200$, Table 2). MLOT and MLVT systems had a number of images approximately three-fold higher than other systems, as they included plots with tree spacings aligned to spacings in SP, CL and VA systems, albeit with leader density maintained constant.

2.3. Fruit number

The Green Atlas protocol involves calibration to obtain absolute fruit number per tree or per hectare. In our case we calibrated detections per image against fruit number per hectare as our experiment included trees at different spacings and a calibration to obtain fruit number / tree might have been misleading due to variable number of trees visible in each image collected using *Cartographer*.

Mobile scans of both sides of short row sections (7–10 m) were used to obtain counts of fruit (detections per image) in the experimental ‘ANP-0131’ orchard at the Tatura SmartFarm. Ground truth counts of fruit were obtained in the same row sections with a commercial fruit grader equipped with optical sensors (InVision 9000; Compac Sorting Equipment Ltd, Collierville, Victoria, Australia) at harvest in years 1 and 2. Ground truth counts were used to calibrate detections per image and predicted counts were expressed in number per hectare (Table 2) to simplify interpretation due to variable tree spacing amongst training systems.

Block-scale mobile scans were carried out to generate fruit number

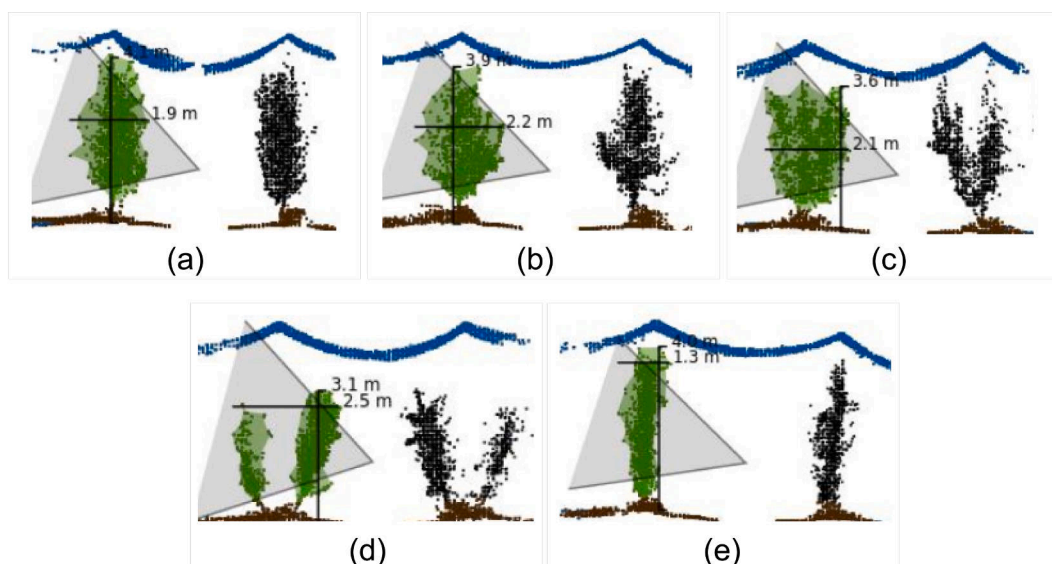


Fig. 1. Two-dimensional LiDAR-derived shape transects of pear trees trained to (a) spindle, (b) central leader, (c) vase, (d) multileader open Tatura trellis, and (e) multileader vertical trellis. The green polygons represent the projected two-dimensional canopy area. The blue and brown points show shade netting and ground points filtered out from canopy geometry calculations, respectively (automatically by the processing software). The vertical and horizontal bars indicate canopy height and canopy width, respectively. Measurements conducted before the 2022 (year 2) fruit harvest.

Table 2

Summary of fruit number scans in 2020–21 (year 1) and 2021–22 (year 2) carried out at the ‘ANP-0131’ experimental orchard at the Tatura SmartFarm. Standard deviations of average images per metre of tree row shown in brackets.

Year	Date of scan	Training system	Randomised blocks	No. of images/ metre of tree row	Total no. of images
1	11 February 2021	SP ^v	3	4.0 (0.3)	506
		CL ^w	3	4.1 (0.4)	515
		VA ^x	3	4.0 (0.2)	502
		MLOT ^y	9	4.2 (0.2)	1590
		MLVT ^z	9	4.0 (0.3)	1515
2	28 February 2022	SP	3	2.1 (0.2)	263
		CL	3	2.0 (0.1)	251
		VA	3	2.1 (0.3)	270
		MLOT	9	2.4 (0.5)	896
		MLVT	9	2.0 (0.1)	753

^v spindle;

^w central leader;

^x vase;

^y multileader open Tatura trellis;

^z multileader vertical trellis.

predictions in the commercial orchard and to obtain estimates of variability. The calibration of fruit number was achieved by fitting a linear regression between manual counts of fruit using click counters and detections per image in four 6-m and seven 7.5-m calibration zones, in the ‘PremP009’ and ‘ANP-0131’ blocks, respectively.

2.4. Peel colour

Peel colour redness was estimated using the colour development index (CDI) derived from hue angle values, as demonstrated in previous work on peach and nectarine (Islam et al., 2022; Scalisi et al., 2022), measured by the mobile platform *Cartographer*. CDI was estimated within the central pixels of the detection boxes generated after deep-learning recognition of pear fruit. CDI conveniently requires no calibration although association with a visual colour scale is helpful for interpretation. Due to cultivar-specific peel colour features, the interpretation of CDI changes between blush pears and fully red pears. While in ‘PremP009’ CDI represents a powerful redness indicator as it is, in ‘ANP-0131’ it was important to unveil its association with blush coverage.

Cartographer scans for CDI were obtained at harvest in year 1 and 2 at the experimental ‘ANP-0131’ orchard, whereas ground truth measurements of blush coverage (%) were obtained after harvest using the fruit grader. A relationship between CDI and blush coverage in ‘ANP-0131’ was established. Normal distributions of blush coverage classes (five histogram bins: 0–20, 20–40, 40–60, 60–80 and 80–100 %) estimated using the commercial grader and predicted using *Cartographer* were compared and classification errors were calculated (% absolute error). The sample obtained with *Cartographer* scans was composed of images, whereas the fruit population measured with the fruit grader was composed of individual fruit. Since in each image, the average CDI of detected fruit is returned, it is expected that the histograms obtained from *Cartographer* data are narrower (i.e., smaller variance) than those obtained on the whole fruit population. Therefore, data of predicted blush coverage histogram was transformed by increasing residuals to determine the lowest error in classifying blush coverage. Three predicted histograms were compared to the observed histogram—(i) an untransformed histogram derived from *Cartographer* data, (ii) a histogram derived from doubling residuals (Eq. (1)), and (iii) a histogram derived from tripling residuals (Eq. (2)).

$$x_1 = \mu + [(\mu - x) \times 2], \quad (1)$$

$$x_2 = \mu + [(\mu - x) \times 3], \quad (2)$$

where x_1 and x_2 represent data transformed by increasing residuals by two and three folds, respectively, μ is the sample mean, and x is the original value. Transformed values below 0 % and above 100 % blush coverage were rounded to 0 and 100 %, respectively.

Validations of CDI predictions in the commercial ‘PremP009’ orchard were carried out in year 2 by comparing *Cartographer* CDI estimates to CDI observations obtained with a portable Bluetooth colourimeter (IDTools, Rowville, Australia) that outputted Cielab/LCh values as the one used in a previous study (Scalisi et al., 2021b) and hue angle was converted to CDI (Scalisi et al., 2022). Ground truth observations of CDI were measured on a sample of 60 fruit per validation zone in four validation zones. In addition, visual colour classes were identified in the four validation zones, although outdoor visual colour classification of ‘PremP009’ fruit is highly subjective due to different shades of orange, red, rust and yellow. Based on their visual fruit peel colour appearance, the four calibration zones were classified as (A) orange–red, (B) red–rust, (C) green–rust, and (D) orange–rust.

2.5. LiDAR-derived measures of canopy geometry

Canopy geometry parameters calculated from LiDAR data included canopy height, canopy area, canopy density and cross-sectional leaf area (CSLA). Canopy area (m²) represented the area of the polygon drawn around the LiDAR-generated points in the scanned transect, excluding the trunk (green polygons in Fig. 1). Canopy density represented the ratio between the number of light beams generated by the LiDAR that bounces back to the light source and the total number of emitted light beams within the canopy area. CSLA was calculated as the product of canopy area and canopy density and is equivalent to the area of the points (comparable to leaves) within the canopy area polygon in the scanned transect, as shown in a previous work (Scalisi et al., 2021a). Canopy geometry parameters were measured before harvest in year 2.

2.6. Relationships between fruit number, peel colour and canopy geometry

Relationships between fruit number, blush coverage and canopy geometry features were investigated in the training systems under study in the ‘ANP-0131’ experimental trial before harvest in year 2.

The relationships obtained in the ‘ANP-0131’ experiment were compared to those obtained in the commercial ‘ANP-0131’ orchard in the same season. To extract calibrated fruit number and blush coverage in the commercial block, a grid of 12 m (3 × row spacing) × 15 m (10 × tree spacing) pseudo-plots was overlaid, as per Scalisi et al. (2023a). In addition, two rows of pseudo-plots along the 4 edges of the orchard block were removed from the crop load relationship analysis as they were considered “buffer” plots that were subject to factors external to the orchard block, as typically done in experimental designs. Relationships in the commercial ‘ANP-0131’ orchard were extracted from a total of 126 pseudo-plots.

2.7. Geoprocessing and statistical analysis

Data at a plot level was extracted by intersecting the points generated by *Cartographer* with plot polygons (i.e., shapefiles), and statistical summaries and colour-coded heatmaps were produced with QGIS (v. 3.16, QGIS Development Team, Open Source Geospatial Foundation, 2021). The geographic precision of plot intersections was improved by real-time kinematic (RTK) positioning adjustments.

Analysis of variance (ANOVA), *t*-tests and linear regression procedures were carried out using R (v. 4.0.2, R Core Team, Vienna, Austria) and its packages “Userfriendlyscience” (Peters 2022) and “DescTools”

(Signorelli 2023). The Games-Howell's posthoc test was used to compare significant differences between groups. The intercept values of linear regression models generated for fruit number calibrations were set to 0. Slope standard errors (SE) and percent standard errors (%SE) were used to measure errors in the fruit number calibrations. Lin's concordance correlation coefficient (r_c) and %SE were used as evaluation metrics for the validation of CDI predictions in 'PremP009'. The root mean square error (RMSE) was used to measure the error of the blush coverage vs CDI relationship. Ghost lines—i.e., “perceptual landmarks” that help integrate the information across graphic panels (Fife et al., 2021)—were used for visual comparison of relationships obtained in the ‘ANP-031’ experimental and commercial orchards. Graphs were generated using SigmaPlot 12.5 (Systat software Inc., Chicago, IL, USA).

3. Results

3.1. Fruit number

3.1.1. Experimental orchard

The relationships between fruit number detections using the mobile platform *Cartographer* and observed fruit number counted using the commercial fruit grader were different in the five training systems under study at the ‘ANP-0131’ experimental orchard at the Tatura SmartFarm (Fig. 2). Different calibration factors (slopes of the linear regressions) were obtained for each training systems. In year 1 (Fig. 2a) and year 2 (Fig. 2b), *Cartographer* detected the highest and lowest number of fruit per unit of ground-truth counts in MLVT and MLOT trees, respectively. When years 1 and 2 were pooled together, the same clear visual separation between detections in MLOT and MLVT systems was measured. In years 1 and 2, detections in VA, SP and CL had similar relationships to observations obtained with the fruit grader.

Linear regression slope comparison carried out with ANOVA tests confirmed visual observations—detections in MLOT and MLVT systems were significantly ($p < 0.05$) the lowest and the highest, respectively, and statistically differed ($p < 0.05$) from each other and from VA, SP and CL (Table 3). In year 1, detections in VA were significantly higher ($p < 0.05$) than SP and CL. In year 2 the same difference was not confirmed. In fact, when years 1 and 2 were pooled together, detections in VA trees were significantly lower ($p < 0.05$) than those in SP trees.

Regression models revealed that, with the exception of SP trees in year 1, calibration had %SE below 6 % in all the training systems (Table 3). In year 1, detections in SP trees had %SE = 10.3 %, although an error above 10 % was not confirmed in year 2. Overall, when years 1 and 2 were considered together, detections in MLVT had calibration errors of 2.2 %, whereas detections in the remaining systems had %SE ranging from 4.0 to 6.4 %.

3.1.2. Commercial orchards

Regression models in commercial orchards with tree canopies trained to Tatura trellis (Y) revealed higher *Cartographer* detection rates per unit of observed fruit number counted with click counters in ‘PremP009’ (slope coefficient = 0.588) than in ‘ANP-0131’ (slope coefficient = 0.497) (Fig. 3).

The calibration errors for ‘PremP009’ and ‘ANP-0131’ were 4 (Fig. 3a) and 1 % (Fig. 3b), respectively; thus, it was confirmed that even in commercial orchard conditions, fruit number detections can be calibrated with errors below 5 %.

3.2. Peel colour

3.2.1. Experimental ‘ANP-0131’ orchard

The scatterplot of the relationship between CDI measurements obtained using *Cartographer* and the blush coverage percentage obtained at harvest with the fruit grader revealed no visual separation between training systems (Fig. 4) and ANOVA comparison between linear slopes showed no statistical separation ($p > 0.05$, data not shown). Thus, a

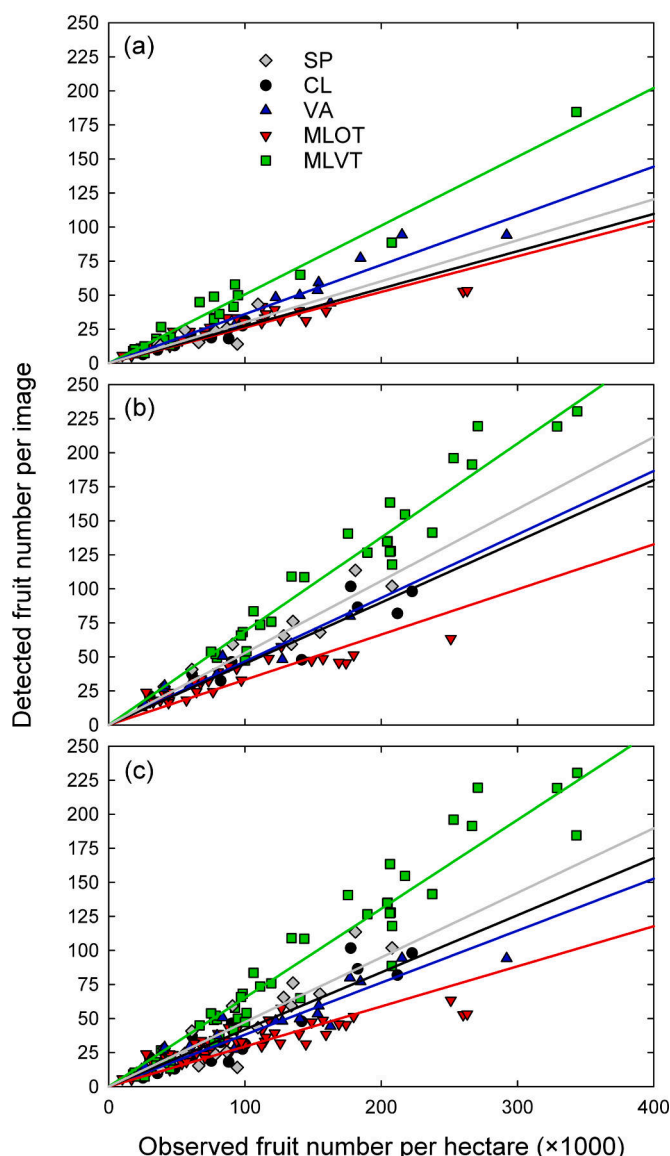


Fig. 2. Linear regression models ($y_0 = 0$) for relationships between fruit number detections using the mobile platform *Cartographer* at harvest and observed fruit number measured with a commercial fruit grader. Measurements carried out in (a) 2020–21 (year 1), (b) 2021–22 (year 2), and (c) year 1 and 2 pooled together. Regression statistics (with intercept set to 0) are shown in Table 3. SP = spindle; CL = central leader; VA = vase; MLOT = multileader open Tatura trellis; and MLVT = multileader vertical trellis.

linear fit was interpolated through the scatterplot points of all the training systems (Fig. 4) and a significant positive relationship ($R^2 = 0.666$, $p < 0.001$) was obtained.

Histograms of blush coverage predicted using the equation in Fig. 4 for years 1 and 2 pooled together showed that untransformed predictions had narrower distribution compared to observations (Fig. 5a). Although the overall predicted mean blush in untransformed predictions and observations was similar (Table 4) and the overall prediction error of average blush coverage in the block was low (< 1.0 %), within-class errors were relatively high—i.e., 40 % error in the 20–40 % blush coverage class, 21 % error in the 0–20 % class and 5 % error in the 40–60 % class, respectively (Fig. 5a). With the transformations carried out by doubling and tripling residuals, mean values remained approximately identical to untransformed data (Table 4). When residuals were tripled, the mean value slightly increased due to the rounding of values < 0 and > 100 % blush coverage, leading to a coincidental match

Table 3

Regression statistics (models with intercept set to 0) for fruit number calibrations at the 'ANP-0131' experimental orchard (Tatura SmartFarm) carried out in 2020–21 (year 1) and 2021–22 (year 2), and combined year 1 and 2. Different letters represent significant differences between regression slopes based on Games-Howell's test.

Year	Training system	Calibration zones	Regression slope	Slope SE ^t	% SE ^u
1	SP ^v	9	0.301 bc	0.0310	10.3
	CL ^w	9	0.274 bc	0.0154	5.6
	VA ^x	9	0.361 b	0.0187	5.2
	MLOT ^y	27	0.262 c	0.0129	4.9
	MLVT ^z	27	0.505 a	0.0141	2.8
2	SP	9	0.529 b	0.0271	5.1
	CL	9	0.450 b	0.0268	5.9
	VA	9	0.466 b	0.0284	6.1
	MLOT	27	0.332 c	0.0175	5.3
	MLVT	27	0.689 a	0.0146	2.1
1 + 2	SP	18	0.474 b	0.0305	6.4
	CL	18	0.419 bc	0.0236	5.6
	VA	18	0.382 c	0.0176	4.6
	MLOT	54	0.294 d	0.0117	4.0
	MLVT	54	0.653 a	0.0146	2.2

^t standard error;

^u percent standard error;

^v spindle;

^w central leader;

^x vase;

^y multileader open Tatura trellis;

^z multileader vertical trellis.

between predicted and observed mean. In addition, tripling the residuals (Fig. 5c) had improved performance on blush coverage predictions compared to doubling them (Fig. 5b), as the within-class errors were significantly reduced. Specifically, the highest error (2 %) was measured when predicting fruit of 60–80 % blush coverage, followed by 20–40 and 40–60 % (1 % error), and errors near 0 % derived from predictions of blush coverage in the 0–20 and 80–100 % classes (Table 4).

3.2.2. Commercial 'PremP009' orchard

The validation of CDI predictions in the commercial 'PremP009' orchard revealed that the ground-based platform *Cartographer* generated estimates of peel redness with good accuracy and precision ($r_c = 0.794$) and %SE = 1.09 (Fig. 6). CDI values also reflected visual classes, as the validation zone with green–rust peel colour had the lowest CDI (0.735), followed by the orange–rust (0.770), red–rust (0.775) and orange–red (0.778) zones.

Standard deviations in predicted CDI were lower than in observed CDI (Fig. 6) due to the sample obtained with *Cartographer* being composed of images, whereas the ground truth sample was composed of individual fruit. Smaller standard deviations of predictions were expected in line with the results on blush coverage distribution of untransformed data shown in Fig. 5a. The ratio between standard deviations of observed values and predicted values resulted in a variance 2.4-fold larger in the former.

3.3. LiDAR-derived measures of canopy geometry

All the canopy geometry parameters obtained in year 2 were significantly affected ($p < 0.001$) by tree architecture (Table 5). Trees trained to MLOT, had the lowest canopy height due to the inclination angle of the V-trellis, but also the largest canopy width considering the two trellis sides. Expectedly, with an average width of 1.5 m, MLVT trees had the narrowest canopies due to their 2D configuration. Canopy area was larger in traditional 3D systems like SP, CL and VA than in MLOT and MLVT, respectively. MLOT trees had the densest canopies, closely followed by MLVT and SP trees. MLVT and CL trees had the lowest cross-sectional leaf areas.

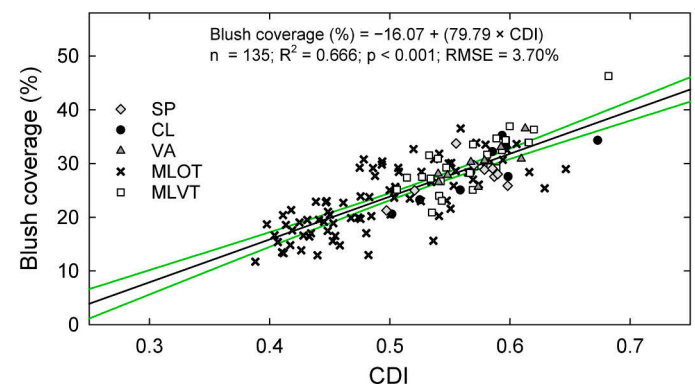


Fig. 4. Linear relationship between blush coverage measured by a commercial grader and colour development index (CDI) estimated using the mobile platform *Cartographer* (Green Atlas, Alexandria, NSW, Australia) in 'ANP-0131' pears in five training systems. Data points represent means of experimental plots in two orchards scanned prior to harvest in the 2020–21 (year 1) and 2021–22 (year 2) seasons. SP = spindle; CL = central leader; VA = vase; MLOT = multileader open Tatura trellis; and MLVT = multileader vertical trellis. The green lines show 95 % confidence intervals.

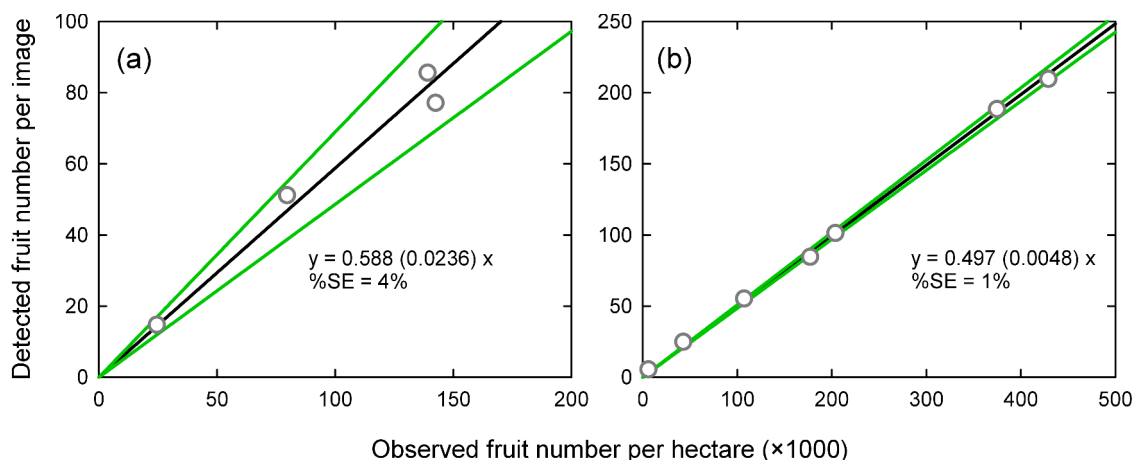


Fig. 3. Linear regression models ($y_0 = 0$) for relationships between fruit number detections using the mobile platform *Cartographer* (Green Atlas, Alexandria, NSW, Australia) at harvest and observed fruit number manually counted in (a) four 6-m sections of the 'PremP009' block, and in (b) seven 7.5-m sections of the 'ANP-0131' block at Calimna Orchard. Slope standard errors reported in brackets and calibration percent standard errors (%SE) reported below the equations.

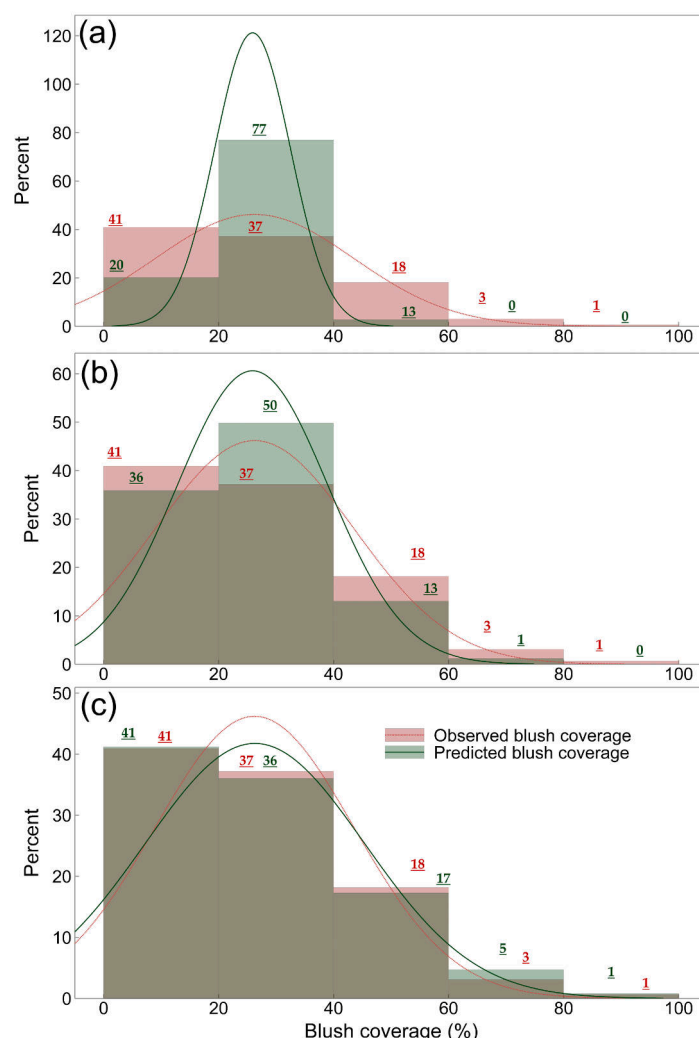


Fig. 5. Five-class histograms showing distributions of blush coverage estimated with a fruit grader (red bars) and predicted using Green Atlas *Cartographer* (green bars) in the experimental ‘ANP-0131’ orchard scanned prior to harvest in the 2020–21 (year 1) and 2021–22 (year 2) seasons. Normal distribution fits are shown as lines—in red for observations, and in green for (a) untransformed predictions, for (b) predictions transformed by doubling residuals, and for (c) predictions transformed by tripling residuals. Numbers in red and green show within-class blush coverage for observations and predictions, respectively.

In 2022, the commercial ‘ANP’0131’ orchard had average (with standard deviations in brackets) canopy height = 3.50 (0.38) m, canopy width = 2.37 (0.42) m, canopy area = 3.24 (0.80) m², canopy density = 0.658 (0.134), and cross-sectional leaf area = 2.20 (0.72) m².

3.4. Relationships between fruit number, peel colour and canopy geometry

3.4.1. Experimental ‘ANP-0131’ orchard

Relationships between fruit number and canopy geometry parameters were established for all the training systems pooled together (Fig. 7), as separating them in different groups deteriorated the quality of the relationships. Fruit number had poorly significant positive correlation with canopy height (Fig. 7a) and negative correlation with canopy width (Fig. 7b) and had no significant correlation with canopy density (Fig. 7d). Fruit number appeared to be subjected to a stronger negative effect of canopy area (Fig. 7c) and cross-sectional leaf area (Fig. 7e).

Blush coverage appeared to be affected by canopy geometry features similarly to how fruit number was affected. Increasing canopy height and width produced significant but poor (i) improvement of blush coverage for the former and (ii) deterioration of blush coverage for the latter (Fig. 7f,g). Canopy area did not significantly affect blush coverage (Fig. 7h). Tree canopies with higher canopy density and cross-sectional

leaf area had a significantly lower blush coverage (Fig. 7i,j).

Overall, in year 2 fruit number per hectare in MLVT trees was the highest, although it did not statistically differ ($p > 0.05$) from CL and SP trees (Fig. 8a). MLOT and VA trees had the lowest fruit number per hectare. When fruit number was expressed per unit of cross-sectional leaf area, differences between training systems remained similar, as MLVT yielded the highest number of fruit per unit of cross-sectional leaf area, whilst MLOT and VA produced the lowest (Fig. 8c).

Only CL had significantly higher blush coverage than MLOT while no other differences were found (Fig. 8b). However, when blush coverage was expressed per unit of cross-sectional leaf area, MLVT had the best performance closely followed by CL trees (Fig. 8d). MLOT trees had the lowest efficiency amongst training systems.

Colour-coded heatmaps in Fig. 9 show the spatial trends of fruit number, blush coverage and cross-sectional area generated using *Cartographer* in the experimental ‘ANP-0131’ orchard, overlaid to a shapefile layer of training system experimental plots. The quantile classification in different colour classes revealed a clear visual separation of fruit number between training systems (Fig. 9a), with datapoints from MLVT experimental plots clearly showing more fruit number (darker colour) than MLOT plots. Similarly, lighter points in Fig. 9b represented reduced blush coverage and were found over MLOT plots, in line with findings shown in Fig. 8, where MLOT trees had the lowest

Table 4

Statistics of blush coverage (%) observed in the fruit population using a fruit grader and predicted (%) in images using the mobile platform Green Atlas *Cartographer* in the 'ANP-0131' experimental orchard (Tatura SmartFarm) in 2020–21 (year 1) and 2021–22 (year 2).

Statistic	Observed blush coverage	Predicted blush coverage I ^x	Predicted blush coverage II ^y	Predicted blush coverage III ^z
Count	85,484	2798	2798	2798
Mean	26.3	25.9	25.9	26.3
S.E. ^r	0.0591	0.124	0.249	0.361
S.D. ^s	17.3	6.58	13.2	19.1
C.V. ^t	65.8	25.4	50.8	72.6
I.Q.R. ^u	25.9	9.13	18.3	27.4
Q1 ^v	12.0	21.0	16.1	11.1
Median	24.0	25.5	25.1	24.7
Q3 ^w	37.9	30.1	34.3	38.5
0–20 class % error	–	21	5	0
20–40 class % error	–	40	13	1
40–60 class % error	–	5	5	1
60–80 class % error	–	3	2	2
80–100 class % error	–	1	1	0

^r standard error;

^s standard deviation;

^t coefficient of variation;

^u interquartile range;

^v first quartile;

^w third quartile;

^x untransformed prediction;

^y predictions transformed by doubling residuals;

^z predictions transformed by tripling residuals.

blush coverage. In addition, a size-coded heatmap of cross-sectional leaf area showed the highest values in correspondence of MLOT plots and the lowest values over the MLVT plots, in line with Table 5.

3.4.2. Commercial 'ANP-0131' orchard

In the commercial 'ANP-0131' orchard, cross-sectional leaf area showed a negative effect on fruit number and blush coverage (Fig. 10), in line with the results shown at the experimental 'ANP-0131' orchard

(Fig. 7i,j). Nevertheless, the ghost line that reproduced the fruit number vs cross-sectional leaf area relationship obtained in the experimental orchard indicated visually different linear fits from the commercial orchard (Fig. 10a). Statistical testing (*t*-test) for differences in slopes and intercepts revealed that the difference between the relationships in the two orchards was significant ($p < 0.001$). Conversely, blush coverage had a similar relationship (slope $p > 0.05$) in the two orchards, as shown by the proximity between ghost line and linear fit (Fig. 10b).

The relationships obtained in the commercial 'ANP-0131' orchard were more robust ($R^2 = 0.536$ and 0.651 for fruit number and blush coverage vs cross-sectional leaf area, respectively) than those obtained in the experimental orchard ($R^2 = 0.340$ and 0.339 for fruit number and blush coverage vs cross-sectional leaf area, respectively).

The commercial orchard had an average fruit number / cross-sectional leaf area = $50.8 (\pm 3.14)$ fruit m^{-2} , which represented a slightly better productive efficiency than MLOT and VA plots, but a lower efficiency than MLVT, SP and CL plots at the experimental orchard in the same season (Fig. 8c). Furthermore, blush coverage / cross-sectional leaf area in the commercial orchard was $10.9 (\pm 0.253)$, which represented reduced efficiency of redness coverage per unit of leaf area compared to MLVT, SP, CL and VA, and better efficiency than MLOT plots at the experimental orchard in the same season (Fig. 8d).

Table 5

Summary results of the analysis of variance for the effects of 'ANP-0131' training systems on canopy geometry parameters measured in year 2 (2021–22). Means and standard errors (in brackets) are reported and different letters represent significant differences ($p < 0.001$) based on the Games-Howell's test.

Parameter	Training system				
	SP ^v	CL ^w	VA ^x	MLOT ^y	MLVT ^z
Canopy height (m)	4.06 (0.07) ab	3.95 (0.14) ab	3.83 (0.20) b	3.31 (0.11) c	4.09 (0.05) a
Canopy width (m)	1.98 (0.14) c	2.04 (0.19) c	2.38 (0.16) b	3.07 (0.22) a	1.53 (0.16) d
Canopy area (m ²)	4.14 (0.40) ab	4.03 (0.51) ab	4.65 (0.49) a	3.92 (0.44) b	3.14 (0.37) c
Canopy density (0–1)	0.60 (0.02) b	0.51 (0.07) c	0.50 (0.04) c	0.67 (0.03) a	0.63 (0.04) b
Cross-sectional leaf area (m ²)	2.49 (0.25) ab	2.10 (0.39) bc	2.40 (0.34) ab	2.61 (0.30) a	1.98 (0.29) c

^v spindle;

^w central leader;

^x vase;

^y multileader open Tatura trellis;

^z multileader vertical trellis.

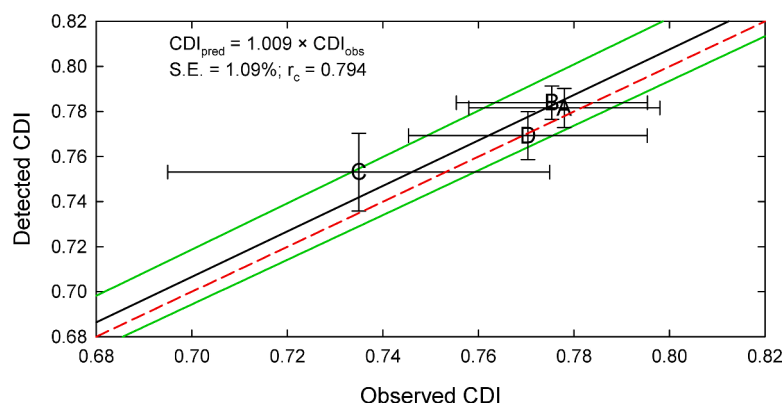


Fig. 6. Linear relationship between colour development index detected using the mobile platform Green Atlas *Cartographer* and colour development index observed with a handheld contact colourimeter in a 'PremP009' commercial orchard (Calimna orchard, Ardmona, Victoria, Australia). Data points represent means of four 6 m zones with horizontal and vertical bars showing standard deviations. The four zones were colour-coded based on visual perception as follows: (A) orange-red, (B) red-rust, (C) green-rust, and (D) orange-rust. The red dashed line shows the 1:1 line, whereas the green lines show 95 % confidence intervals. Fruit sample size for observed CDI = 60; image sample size for detected CDI = 13.

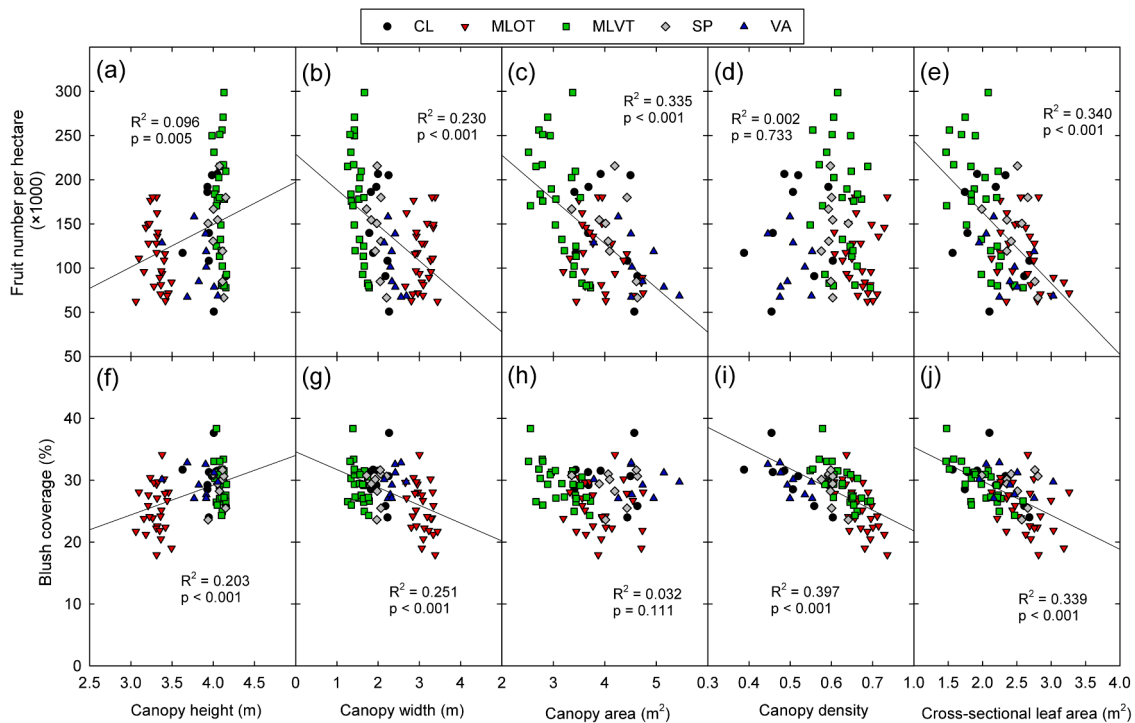


Fig. 7. Blush coverage and fruit number linear relationships with canopy geometry parameters—i.e., (a, f) canopy height, (b, g) canopy width, (c, h) canopy area, (d, i) canopy density, and (e, j) cross-sectional leaf area—in different training systems. Data shows relationships at the ‘ANP-0131’ experimental site in 2021–22 (year 2). Coefficients of determination (R^2) and p-values of the linear regressions reported in each panel. SP = spindle; CL = central leader; VA = vase; MLOT = multileader open Tatura trellis; and MLVT = multileader vertical trellis.

The heatmap in Fig. 11 shows colour-coded points (light red to dark red) representing spatial variability of blush coverage overlaid to size-coded points highlighting the spatial variability of cross-sectional leaf area at the ‘ANP-0131’ commercial orchard. The map provides a visual representation of the inverse relationship between blush coverage and cross-sectional leaf area, with orchard zones with less developed foliage having the highest blush coverage.

4. Discussion

The scientific validation of technology is a prerequisite to allow reliable upscaling and trustworthy integration with front-end management automation. Such a scalable approach will boost orchard digitisation and build historical databases to support long-term sustainability of orchard businesses.

In this study, it was demonstrated that the number of fruit detections per unit of ground-truth counts is lower in denser canopy configurations (3D) compared to narrow vertical 2D systems (Fig. 2, Table 3). This is likely due to fruit being increasingly occluded by foliage in visible portions of the canopy (Koiralet al., 2021; Tang et al., 2023). Nevertheless, when considering the two years of study, predictions errors after applying a calibration factor remained low in all the 3D architectures—range between 4.0 and 6.4 % (Table 3). In the same time range, errors in MLVT trees were below 2.5 %. Crop load prediction errors remained below 5 % in commercial orchard tests carried out in year 2 (Fig. 3). Similarly, fruit detections in stone fruit canopies previously showed better machine vision performance in trees trained on trellises, as opposed to trees with free standing large canopies (Islam et al., 2022). Improved performance of machine vision on 2D canopy systems was also measured in apples (Bortolotti et al., 2021).

There is a worldwide trend for orchard intensification that has seen a significant increase of pear high density plantings (Musacchi et al., 2021; Sansavini and Musacchi, 2002). Narrow-wall, multileader training systems have shown great potential to increase productivity and

fruit quality (Dorigoni and Micheli, 2018), as trees with large canopies produce fruit with highly variable maturity and quality (Zhang et al., 2016). While increasing orchard intensification is not primarily driven by improved efficiency of automation, robotics and technology overall—rather by factors such as labour cost (Musacchi et al., 2021), spray volume (Sansavini et al., 2008), and packout (Dorigoni et al., 2011)—the improved machine vision performance for crop load estimations in 2D systems provides indications that these systems are the most suitable tree architectures for accurate future orchard digitising and phenotyping, and precision horticulture applications.

Despite different canopy geometry features in the training systems under study (Table 5), estimates of blush coverage derived from CDI were consistent in SP, CL, VA, MLOT and MLVT trees (Fig. 4). This is likely because measurements of CDI were carried out early in the morning and the strobe lights on *Cartographer* that continuously flashed canopies guaranteed a standardisation of the colour appearance and light environments under diverse canopy conditions. Blush coverage is a very valuable parameter for ‘ANP-0131’ pears in the Australian context. Similarly, CDI predictions in a commercial ‘PremP009’ block had good accuracy and precision ($r_c = 0.794$, %SE = 1.09 %) in a scenario in which operators visually assessing peel colour failed to clearly identify substantial differences between zones.

Although block- or zone-scale blush coverage averages have good accuracy (Figs. 4 and 6), classification of fruit into blush coverage histogram classes using untransformed data revealed failure to represent an accurate spread of data (Fig. 5a). In detail, the errors for the percentage of fruit with blush coverage in the 20–40 and 0–20 % classes were 40 %. Nonetheless, transformation of data by tripling the residuals of the raw data revealed a much improved performance, with within-class errors reduced to ≤ 2 % (Fig. 5c, Table 4). This was expected, as the sample obtained with *Cartographer* scans was composed of images, whereas the fruit population measured with the fruit grader was composed of individual fruit.

Obtaining accurate block estimates of CDI (and derived blush

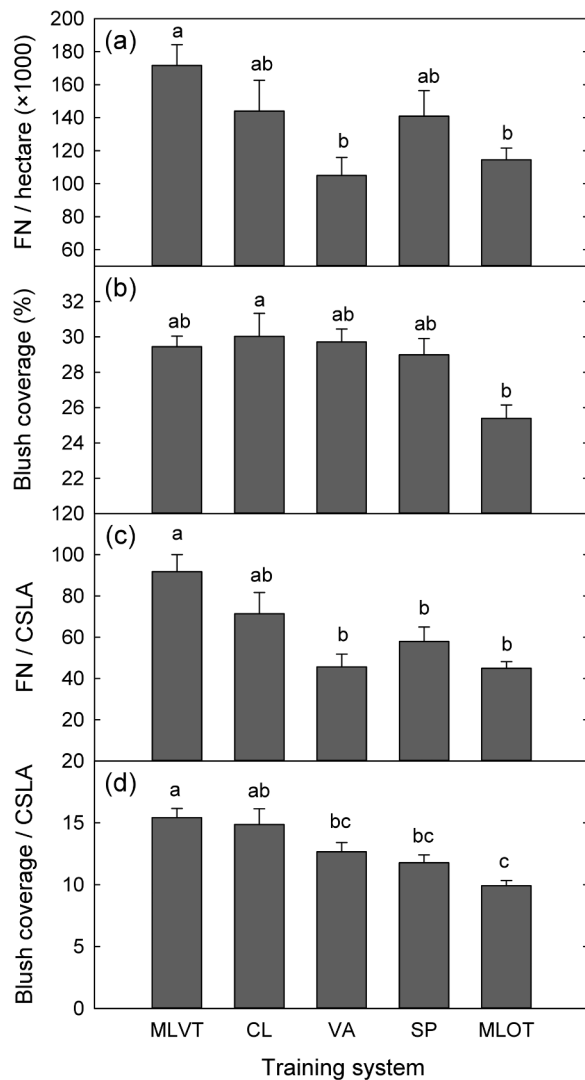


Fig. 8. Bar charts for (a) fruit number (FN), (b) blush coverage, (c) FN per unit of cross-sectional leaf area (CSLA) and (d) blush coverage per unit of CSLA in five training systems—SP = spindle; CL = central leader; VA = vase; MLOT = multileader open Tatura trellis; and MLVT = multileader vertical trellis. Different letters within each panel represent significant differences ($p < 0.001$) following analysis of variance and Games-Howell's test; error bars show standard errors of the means.

coverage for blush cultivars) in advance of harvest can help growers manage colour development (e.g., reflective mulching, retractable netting, biostimulants) in the orchard to improve packout and inform the supply chain in a timely manner (Scalisi et al., 2022). The availability of temporal data can also support harvest logistics and reduce waste. It is equally important to be able to accurately predict peel colour histogram classes, as shown for fruit size in recent studies (Apolo-Apolo et al., 2020; Gongal et al., 2018; Scalisi et al., 2023b). While several research studies have focused on large-scale crop load and yield predictions, only few attempts have been done to estimate the spatial variability of peel colour in orchards (Scalisi et al., 2022). While hand-held colourimeters or operator visual assessments remain the most adopted methods to assess peel colour *in situ* (Scalisi et al., 2021b), machine vision has great potential to upscale data collection on much larger sample sizes, although its use is still limited due to the significant effects of external light on pixel colour (Mirbod et al., 2021; Silwal et al., 2021). The findings obtained in this study provide confidence on the reliability of ground-based platforms for large-scale estimations of peel colour.

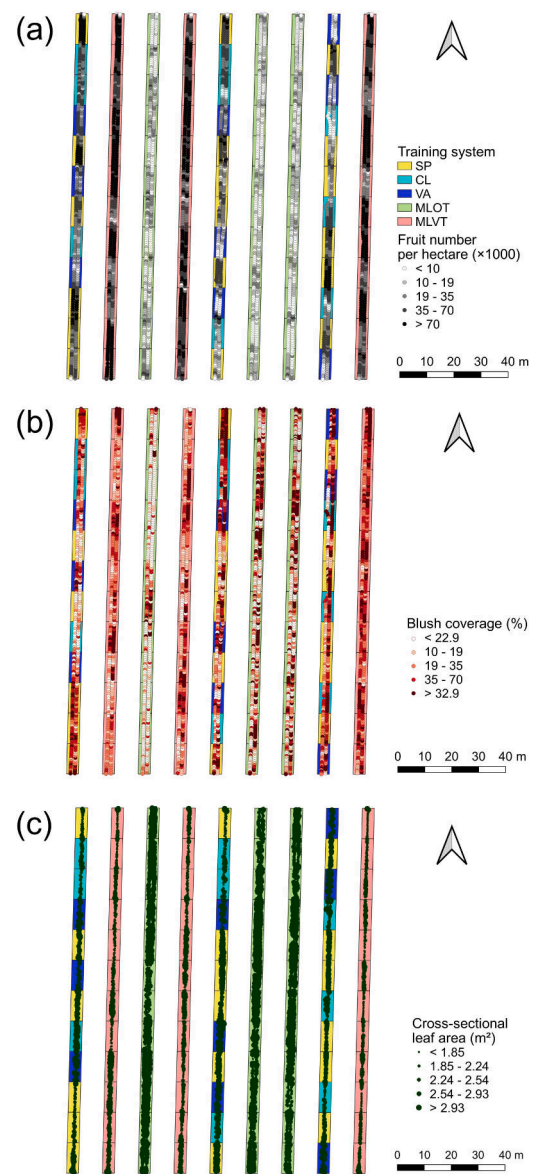


Fig. 9. Heatmaps of (a) calibrated fruit number, (b) blush coverage, and (c) cross-sectional leaf area in the experimental 'ANP-0131' pear orchard located at the Tatura SmartFarm. Data points represent images generated with the mobile platform Cartographer (Green Atlas) before the 2022 harvest. Experimental plots with different training systems shown in rectangles with different colours—SP = spindle; CL = central leader; VA = vase; MLOT = multileader open Tatura trellis; and MLVT = multileader vertical trellis.

The canopy geometry parameters obtained from LiDAR images generated from Cartographer returned valuable indicators to characterise tree architectures in different pear training systems (Table 5). Obtaining key canopy features is essential to precisely manage or automate operations such as variable-rate spraying (Escola et al., 2013; Li et al., 2018; Seol et al., 2022; Sultan Mahmud et al., 2021), pruning and defoliation (Westling et al., 2021), and irrigation (Escola et al., 2017), as point clouds may be used to derive variable crop coefficients that can be incorporated in crop water requirement models and precision irrigation scheduling. In addition to practical management operations, LiDAR cloud points have been used to model or estimate light interception (Örn, 2016; Scalisi et al., 2021a; Westling et al., 2018). However, further studies are needed to estimate within-tree light distribution at lateral shoot levels or near fruit to achieve improved fruit quality.

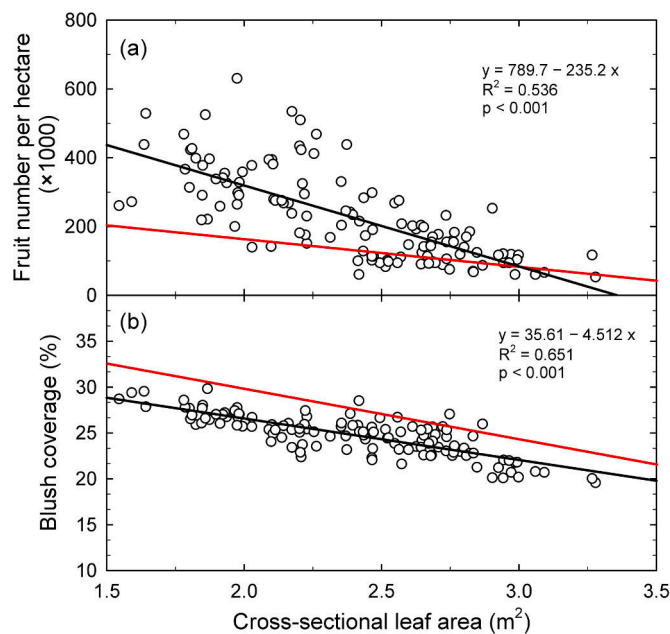


Fig. 10. Scatterplots of (a) fruit number and (b) blush coverage against cross-sectional leaf area obtained in 2021–22 (year 2) at the commercial ‘ANP-0131’ orchard (Calimna orchard, Ardmona, Victoria, Australia) before harvest. Linear fits (black lines), equations, coefficients of determination (R^2) and p-values of the linear regressions are reported in each panel. Ghost lines, representing the linear fits obtained in the experimental ‘ANP-0131’ orchard at the Tatura SmartFarm, are shown in red.

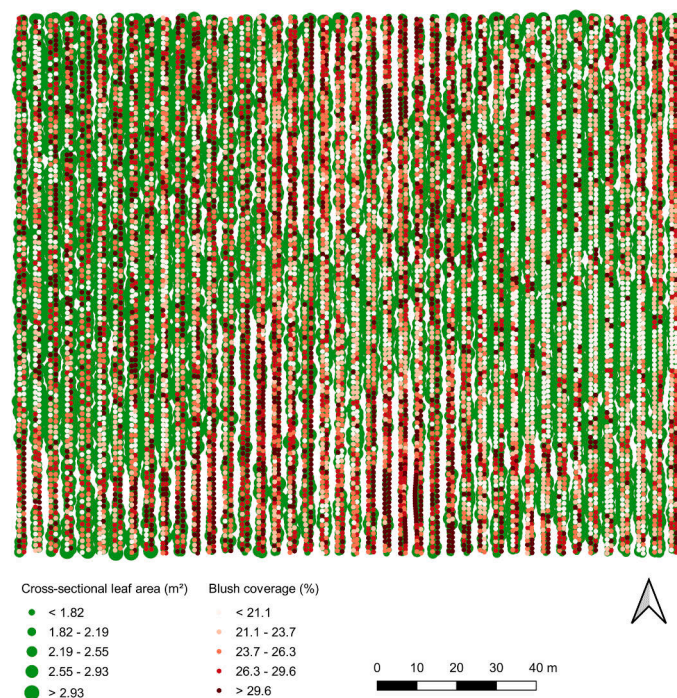


Fig. 11. Heatmap of cross-sectional leaf area and blush coverage in the commercial ‘ANP-0131’ orchard (Calimna orchard, Ardmona, Victoria, Australia). Data points represent images generated with the mobile platform *Cartographer* (Green Atlas) before the 2022 harvest.

Fruit number and blush coverage were both influenced by canopy geometry features (Fig. 7) when data from different canopy architectures were pooled together. It would be better to separate training systems to develop independent relationships between canopy features and

crop load and quality, although datasets should be collected over several years to remove seasonal effects as in this study no separation was recommendable based on visual inference. The negative effects of canopy width, area and cross-sectional leaf area on fruit number (Fig. 7b,c, e) might have been caused by reduced light penetration that led to low bud initiation. In addition, low fruit number might have resulted in an overall increased vigour. These findings suggest that vegetative growth and reproductive (fruit) activity might have been unbalanced towards the former, and improved pruning management strategies will need to be implemented (Musacchi, 2021). The detrimental effects of increasing canopy density and cross-sectional leaf area observed on the blush coverage of ‘ANP-0131’ (Fig. 7i,j) were consistent with previous findings that showed negative effects of reduced exposure of fruit to light on colour formation in the same cultivar (Peavey et al., 2020), in ‘ANP-0118’ (Peavey et al., 2020, 2022), and in ‘d’Anjou’ (Zhang et al., 2016). High shading and poor light environment in internal positions of the canopy is well known to cause reduced colour formation and fruit quality in general. In this study, it was observed that MLVT trees had the highest fruit number and blush coverage per unit of cross-sectional leaf area, indicating that they could be exposed to an optimal light environment. Due to their 2D configuration, as seen above, MLVT trees are the most suited for the application of robotics, but it is important to consider that, in some cases and in the first years after planting, the management of these canopies might require more specialised work and higher costs compared to traditional large canopy systems (Musacchi et al., 2021).

Findings on negative effects of cross-sectional leaf area on fruit number and blush coverage were confirmed in a commercial orchard (Fig. 10). In this case the effects were more robust as the orchard was larger and not characterised by a pool of different architectures. In addition, the relationships between canopy size, fruit number and blush coverage in the experimental ‘ANP-0131’ trees were influenced by overhead netting, as opposed to unnetted commercial trees. Georeferenced colour- and size-coded maps (Fig. 11) provide a valuable tool to visualise overlaid canopy information and fruit parameters (e.g., number and colour) to easily identify zones that necessitate priority interventions and to tailor precision management practices to specific requirements.

Advanced phenotyping of tree architectures can be achieved by integrating LiDAR and machine vision (Huang et al., 2020; Rosell and Sanz, 2012; Underwood et al., 2016). The integration of LiDAR and machine vision has important implications for breeding programs. Their combination supports the characterisation of selections that are best suited for 2D planting and for phenotyping key tree architectural and reproductive traits for pear production systems, such as branch angle, fruit-leaf ratio, fruiting behaviour, fruit clustering, as summarised in detail in available literature (Costes et al., 2004, 2010; Lauri et al., 2002; Musacchi, 2021).

5. Conclusions

The platform used in this study had consistent and reliable performance, placing it in an optimal position for scientific data acquisition in research experiments, while maintaining its prerogative of being a commercially valuable tool for growers to measure and map spatial information and digitise fruit and tree metrics. Digitising orchards and crop characteristics is a required step towards automation and data-driven precision farming. Building accurate historical datasets of orchard blocks and farms will allow a better understanding of the business by enabling financial analysis and sustainability studies over time based on objective data. The implications of digitised orchards for the automation of orchard practices, from planting to managing and harvesting, are vast and require vertical integration of follow-up technologies that can take data-driven decisions to optimise production. Data predicting packout, based on fruit number, size and colour, is highly sought by the supply chain and could be leveraged by growers and packhouses to

improve supply contracts and management of storage logistics, and in turn generate a positive effect on business profitability.

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Financial interests

James Underwood, Steve Scheduling and Peter Morton, in their capacity of co-founders of Green Atlas, have a commercial interest in the research and development of Cartographer. Alessio Scalisi, Lexie McClymont and Ian Goodwin declare that they have no competing financial or non-financial interests, or personal relationships that could have appeared to influence the work reported in this paper. Agriculture Victoria has conducted independent research and we declare that findings are genuine. The funders had no role in the design of the study, in the collection, analyses or interpretation of data, and in the writing of the manuscript or in the decision to publish the results.

CRedit authorship contribution statement

Alessio Scalisi: Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation, Writing – original draft, Visualization. **Lexie McClymont:** Conceptualization, Methodology, Formal analysis, Data curation, Writing – review & editing. **Maddy Peavey:** Methodology, Writing – review & editing, Resources, Visualization. **Peter Morton:** Software, Resources, Writing – review & editing. **Steve Scheduling:** Software, Resources, Writing – review & editing. **James Underwood:** Software, Resources, Writing – review & editing. **Ian Goodwin:** Conceptualization, Methodology, Resources, Writing – review & editing, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Alessio Scalisi reports financial support was provided by Horticulture Innovation Australia Ltd. James Underwood, Steve Scheduling and Peter Morton report a relationship with Green Atlas that includes: board membership, employment, and equity or stocks.

Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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